

Runtime architecture and operating procedures

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Structured companion: runtime capital source, implementation_assumptions

Architecture decisions

ADR-01: retain Switch Reno as the proposed transitional primary, independently administered IDACORE Boise recovery, and a separate immutable/offline-copy boundary. The owned Northern Nevada site is acquired/preconstruction in the synthetic case, with no vertical construction, completed shell or commissioned plant. Its first planned critical module is 250 kW, with a second 250 kW module conditional on demand, utility design and funding. A 1 MW utility path and optional 2 MW route are design requirements, not energized capacity.

ADR-02: use a small supported platform foundation: Linux/KVM hosts for separate trust domains; containerized application processes inside those domains; PostgreSQL for transactions and temporal metadata; customer-controlled object storage for immutable source artifacts and backups; rebuildable search indexes; a dedicated identity/PAM service and HSM-backed keys. Do not introduce a second authoritative transaction store or let search indexes become the source of truth. PostgreSQL's [published version policy](#) provides a five-year major-version support model. Major 17 is the provisional database baseline; supported point releases, exact OS/orchestrator/object-store versions and signed images require a qualified bill of materials before deployment. The model is deliberately not represented as a tested product integration.

ADR-03: separate enterprise shared services, Alexandria restricted records, Atlas professional services, Atlas client tenants, Foundry central support/control, development/evaluation, management/OOB, backup vault and recovery bootstrap. Business product engineering retains product authority. Provider facility systems cannot administer these planes. Production identities cannot administer backup retention. Customer-controlled edge devices terminate transport; authenticated management is isolated from application and provider networks.

ADR-04: workload-driven sizing replaces the inherited 75/85/120 and later kW envelopes as an executable design sensitivity. Inputs include client hosting share, active users, requests, CPU-seconds, memory, tokens/task, peak windows, model weight/KV memory, accepted throughput assumptions, storage growth and protection copies. Throughput and equipment costs remain synthetic configuration classes. No benchmark, probability or supplier quote is implied. One host/system spare per class and a 70% utilization ceiling are separate resilience and operating reserves. A/B feeds do not double productive demand. Rack-space, power and weight limits are enforced per rack by a deterministic placement model. The base Reno design requires eleven racks and Boise three. Thermal, connector and vendor acceptance remain distinct from this arithmetic.

ADR-05: the 2027 base case estimates Reno 96.4 kW peak, Boise 23.6 kW peak, with typical synthetic draw 60.05/14.6 kW. Boise carries all 150 TiB durable data and reduced compute/accelerator throughput. Search and basic records do not require full Reno AI throughput. The inherited 25 kW envelope is adequate only for that configuration-class arithmetic; later base demand needs expansion. A 10 Gbps network cannot restore all 150 TiB within eight hours. Pre-positioned validated copies, transaction logs, independent keys and bootstrap are mandatory prerequisites.

Authorization contract and bounded disclosure

Every protected resource carries identity, tenant, source/license rights, permitted purposes, information class, compartments, source restrictions, detail/existence readers, expiry, restriction owner/reason, review and challenge route. Every request has an authenticated principal, scoped purpose, action and evaluation time. Missing protected rights deny detail. Open internal questions remain broadly visible. Rank does not grant access; an entitled new ARU hire must not suffer unjustified denial. Legal/source restrictions control whether even existence can be disclosed.

Apply policy before search results, snippets, counts, graph edges, citations, vectors, exports, tool output and model context. Protected institutional discovery can execute in a separate service boundary, but the user's model receives only authorized context. An approved existence signal contains no title, identity, quantity or protected inference. Aggregation must evaluate its complete source lineage, suppressing any unentitled contribution before output. A model cannot be the sole concealment mechanism after restricted text enters its context.

Daedalus cannot write authoritative records, promote its output or acquire institutional authority. Persistent personal memory is segregated by user; shared skills/templates carry no memory, extracts or conversations. No private common-model training is authorized. Agent delegation narrows rights, tenant and purpose. Declassification requires a named information owner, Legal/source-rights review where applicable, reason, scope, authorized recipients, expiry and human record; the reference has no automatic declassification path.

`security.py` tests this contract with synthetic identities. It is not a universal semantic leakage detector or deployed authorization service. Protected payloads are excluded before the reference disclosure path, including prompt-injection strings. Additional deployed-model red teaming and access-inference evaluation remain production gates.

Retention, holds and restore suppression

Records/Legal and information owners approve class-specific obligations; Technology implements them. Proposed operational/security logs: 90 days searchable plus 275 days restricted archive, subject to incident preservation and lawful limits. Temporary model contexts: session-bound; no persistent reuse without explicit user selection and rights. Personal workspace artifacts: user-controlled with periodic rights review. Authoritative Canon and source-protection records follow their own approved schedules; operational-log TTLs do not govern them. Raw collection is purpose/rights-limited, not an indefinite event hoard. These periods are delegated design proposals, not statements of universal legal requirements.

Deletion/revocation creates a versioned tombstone, invalidates caches, indexes, replicas and derived outputs, and suppresses restoration from backups. Legal hold retains protected bytes in a restricted preservation boundary while disabling ordinary disclosure. Backups expire under their approved schedule; a restore must replay current holds, tombstones and entitlements before serving traffic. Keep minimal lawful lineage/timing without retaining prohibited content. Record the affected population, job results, exceptions, owner and review; failed propagation is an incident. `revoke_graph` and `restore` exercise this behavior, not real erasure.

RT-OPS-01

Installation and asset acceptance

Prerequisites: accepted order/configuration, qualified personnel, asset list, maintenance window, customer work authorization and rollback. Receiving records serials, seals, custody and quarantine. Platform compares hardware/firmware to the approved BOM; Security validates baseline, segmentation and management paths. Procurement reconciles delivered quantities; Finance records asset class and readiness separately from payment. Execute approved load and loss-of-component tests, log results and obtain service-owner acceptance. Failed checks retain quarantine and revert to the prior configuration; no production label is granted.

RT-OPS-02

Change, patch and rollback

Owner opens a traceable change with affected services, risk, tests, rollback, entitlements and window. A qualified peer reviews immutable artifact hashes and configuration; emergency authority is scoped and retrospectively

reviewed. Deploy to a representative isolated environment, verify health and recovery, then stage rollout. Check monitoring, transaction integrity and denied-access cases. On breach of acceptance limits, restore the last accepted artifact and reconcile transactions; escalate unresolved data changes. Preserve approvals, diffs, execution identity, timestamps, outcomes and rollback evidence. Monthly lifecycle review identifies unsupported software, vulnerabilities and exceptions.

RT-OPS-03

Access and key lifecycle

IAM accepts authoritative joiner/mover/leaver/service requests and named owners. Provision only scoped rights; separate privileged and break-glass identities. Key custodians perform documented dual-control ceremonies and independent recovery copies; never place real secrets in this repository. Quarterly and event-driven reviews reconcile complete HR/vendor/identity populations. Test revocation against application, cache and recovery paths. On unauthorized access revoke sessions, preserve evidence and invoke Security incident command. Retain request, approval, inventory, rotation/revocation proof and reviewer outcome.

RT-OPS-04

Incident, provider maintenance and environmental alarms

Service owner or Security declares severity and incident commander; authority is temporary and explicit. Record discovery, impact, affected dependencies and notification deadlines. Coordinate provider access, customer notices and safe containment. A temperature/power alarm, lost carrier or log pipeline failure creates an incident even if service remains available. Never suppress loss of redundancy because SLA credit is zero. Validate restored service, logging and data integrity before closure. Preserve complete tickets, telemetry and decision timestamps; track corrective actions and independent retest.

RT-OPS-05

Failed backup and independent restore

Prerequisites: accepted recovery scope, off-primary runbooks/artifacts, independent identity/DNS/time/key access, protected backup and transaction-log populations, qualified on-call roles and authorized test window. Failed or missing backup jobs alert Reliability, reconcile expected versus observed jobs, preserve prior copies and retry under change control. Repeated failures escalate; replication is not accepted as a substitute for immutable/offline copies.

Restore bootstrap first (target one hour, no loss of acknowledged key changes), then prioritized production (four hours/15 minutes), Alexandria (eight hours/one hour), and rebuildable development (24 hours/24 hours). Check each prerequisite, recovery-point cutoff and integrity before starting dependent services. Record start/stop clocks, excluded/degraded features, volume and failures. Rebuild indexes from verified stores. Failback quiesces/diverges safely, reconciles transaction identifiers and integrity, and retains a rollback copy before switching routing. Missed objectives remain failed tests; no calendar or synthetic PASS verifies production recovery. Annual integrated exercise and quarterly component tests are the proposed cadence, with actual performance evidence outstanding.

RT-OPS-06

Capacity, hardware replacement, media and exit

Monthly review compares measured load and available failure capacity with workload drivers, lead times and funding. Approve expansion only with provider delivery, architecture, recovery and Finance evidence. Hardware replacement uses authorized remote hands, asset identifiers, custody and verified rebuild; unexpected data or seal/serial discrepancies stop work. Encrypt media under customer keys; return, reuse or destruction requires owner authorization and recorded custody/method. Exit rehearses data/key portability, replacement operation, de-installation, transport, access revocation and final financial/evidence reconciliation. No provider may receive customer keys as an equipment-release condition.

Evidence form common to procedures

Record procedure/version, boundary, actor/role, authorizer, effective/recorded times, prerequisites, source population, exact configuration/hash, expected and observed results, exceptions, rollback, retained artifact locations, reviewer and next action. Empty or synthetic evidence stays explicitly classified. Management owns execution and remediation; Internal Audit does not operate these controls.

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